Exponential and quadratic functions

1

a Function A

b Function A is decreasing and function B is increasing.

2

a $f(x)$

b $g(x)$

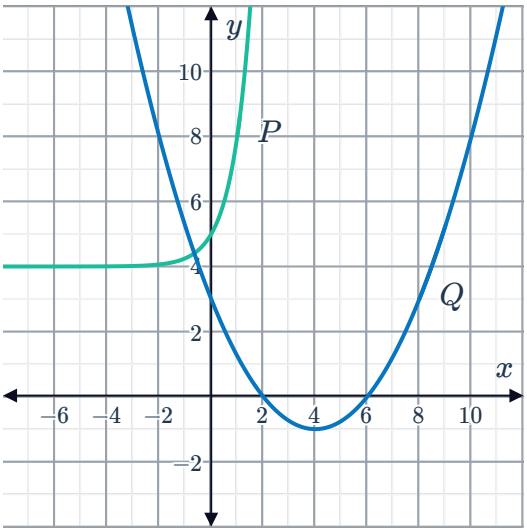
3

a $g(x)$

b Both

4

a



b P

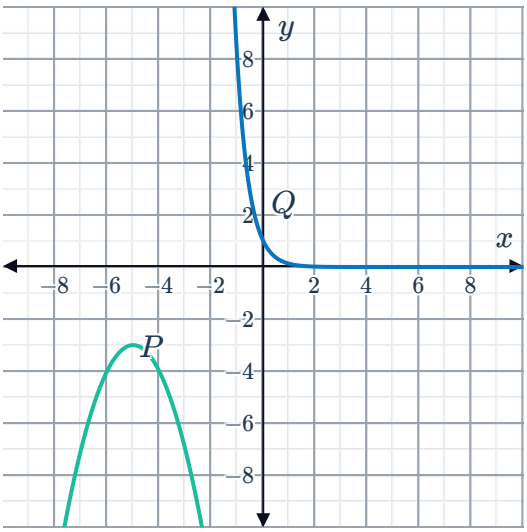
c Function P is increasing and function Q is decreasing.

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5

a Function P is quadratic (parabola) and function Q is exponential.

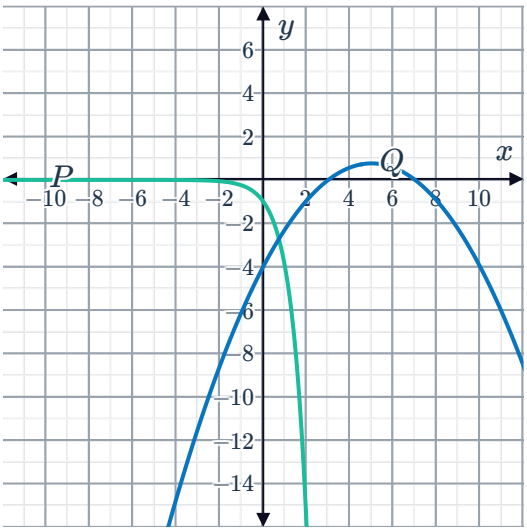
b



c Zero times

6

a



b Once

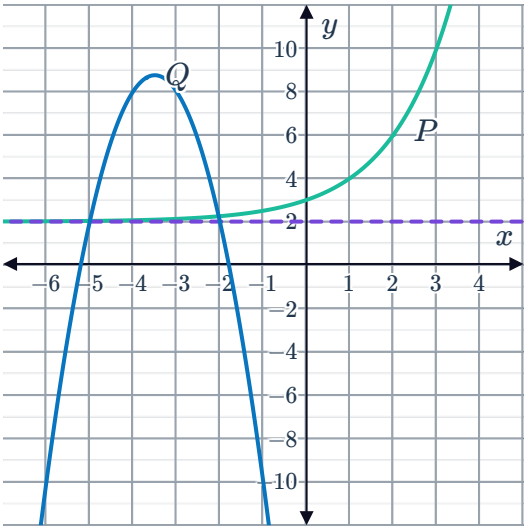
c P

d Function P is decreasing and function Q is increasing.

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7

a



b As x approaches ∞, the function Q approaches $-\infty$ while function P approaches ∞.

c 2

8

a

x	$f(x)$	Increase in $f(x)$	$g(x)$	Increase in $g(x)$
0	1	–	0	–
1	2	1	2	2
2	4	2	8	6
3	8	4	18	10
4	16	8	32	14
5	32	16	50	18
6	64	32	72	22
7	128	64	98	26

b The quadratic function increases more rapidly at first, but then the exponential function starts to increase more rapidly.

9

a The exponential function's increase over a 1 unit interval of x is greater than the quadratic function's increase over the same interval.

b The gap between the functions becomes greater and greater.

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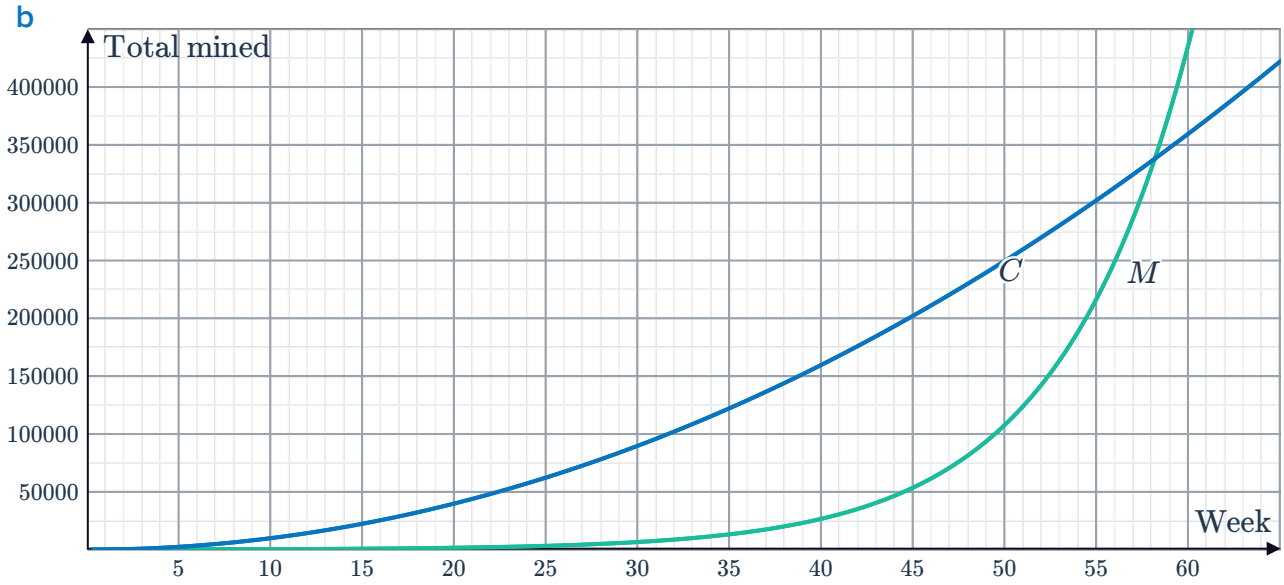
Applications

10

a

n	M	C
1	100	100
5	175	352
10	708	2500
15	10 000	22 500

b



c Crest Corporation

d $(1, 100)$ and $(59.5, 353563.1)$, at the end of the first week of operations the two companies have mined the same total of 100 metric tons. The Crest Corporation then has a greater total amount mined until the approximately 59.5 weeks later when they are equal again at 353563.1 metric tons mined in total.

e Mint Corporation

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11

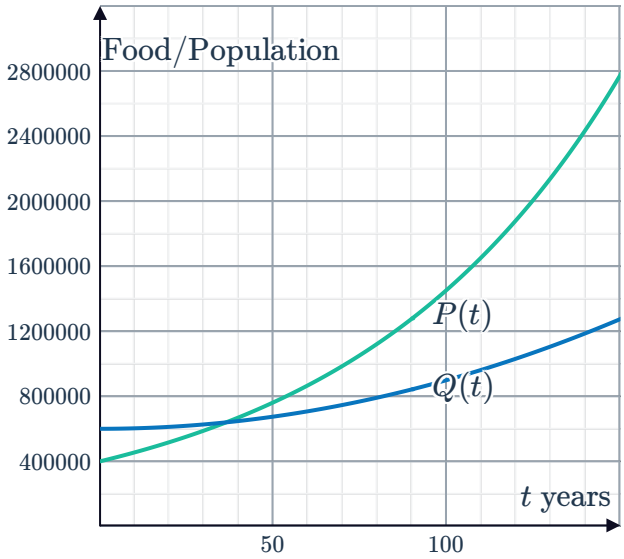
a $147\,850$

b There will be a surplus of food in 10 years time.

c $-555\,474$

d There will be a food shortage in 100 years time.

e



f 36.34 years

g No, the population increases at a faster rate than the food supply. Once the population has overtaken it will stay above and there will be an increasing gap.

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